

HW #3 Solutions

1. Let $X = \text{span}\{S_1 \cup S_2\}$ and $Y = \text{span}\{S_1\} + \text{span}\{S_2\}$. We need to show that $X \subset Y$ and $Y \subset X$, because this is equivalent to $X = Y$. We denote the field by F .

Claim 1 $X \subset Y$.

Proof Let $x \in \text{span}\{S_1 \cup S_2\}$. Then $\exists v_1, \dots, v_k \in S_1$ and $w_1, \dots, w_m \in S_2$, as well as $\alpha_1, \dots, \alpha_k \in F$ and $\beta_1, \dots, \beta_m \in F$ such that $x = \sum_{i=1}^k \alpha_i v_i + \sum_{i=1}^m \beta_i w_i$.

We observe ~~that~~, by definition of $\text{span}\{S_1\}$ and $\text{span}\{S_2\}$, that

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$$\sum_{i=1}^k \alpha_i v_i \in \text{span}\{S_1\} \quad \text{and} \quad \sum_{i=1}^m \beta_i w_i \in \text{span}\{S_2\}$$

Therefore, by the definition of the sum of two subspaces,

$$x = \sum_{i=1}^k \alpha_i v_i + \sum_{i=1}^m \beta_i w_i \in \underbrace{\text{span}\{S_1\} + \text{span}\{S_2\}}_{\text{span}\{S_1 + S_2\}}$$

□

Claim 2 : $Y \subset X$.

Proof: $y \in Y$ means there exists

$y_1 \in \text{span}\{S_1\}$ and $y_2 \in \text{span}\{S_2\}$

such that $y = y_1 + y_2$.

$y_1 \in \text{span}\{S_1\}$ implies there exist

$\bar{\alpha}_1, \dots, \bar{\alpha}_k \in \mathbb{F}$ and $\bar{v}_1, \dots, \bar{v}_k \in S_1$ such that

$$y_1 = \sum_{i=1}^k \bar{\alpha}_i \bar{v}_i.$$

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Similarly, there exist $\bar{\beta}_1, \dots, \bar{\beta}_m \in \mathcal{F}$

and $\bar{w}_1, \dots, \bar{w}_m \in S_2$ such that

$$y_2 = \sum_{i=1}^m \bar{\beta}_i \bar{w}_i.$$

Therefore $y = y_1 + y_2$ is a linear combination of elements of $S_1 \cup S_2$ and thus $y \in \text{span}\{S_1 \cup S_2\} = \mathcal{X}$. □

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2.

(a) Linearly independent if, and only if,
the only solution to

$$\alpha_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

is $\alpha_1 = \alpha_2 = \alpha_3 = 0$. The above

is equivalent to

$$\underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 5 \\ 3 & 0 & 9 \end{bmatrix}}_{\det = 0} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

In deed $3 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

($\alpha_1 = 3, \alpha_2 = -1, \alpha_3 = -1$), and thus
the set is linearly dependent.

(b) $\mathbb{R}^3$ has dimension 3, and thus any set of 4 vectors must be linearly dependent. We do not need to do any further calculations.

$$(c). \quad \alpha_1 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

if, and only if,

$$\underbrace{\begin{bmatrix} 3 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{\det=1 \neq 0} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

∴ Only solution is $\alpha_1 = \alpha_2 = \alpha_3 = 0$

and the vectors are linearly independent.

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Q.3. Suppose that

$$\alpha \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} + \beta \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} + \gamma \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

This is equivalent to

$$\alpha + 2\beta + 4\gamma = 0$$

$$2\alpha + \beta - \gamma = 0$$

$$\alpha + \beta + \gamma = 0$$

(There are only 3 equations because the matrices are symmetric).

Hence

$$\underbrace{\begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}}_{\det = 0} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence, there will exist a nontrivial solution. Indeed, one solution is

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}.$$

Hence, the set is linearly dependent.

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#4 Claim. Let $Y \subset X$ be a subspace and $S \subset Y$. Then $\text{Span}\{S\} \subset Y$.

Proof: From the definition of span, if $S_1 \subset S_2$, then $\text{Span}\{S_1\} \subset \text{Span}\{S_2\}$. Also, if Y is a subspace, then $\text{Span}\{Y\} = Y$. Putting these two observations together we have

~~span~~ $S \subset Y$, Y a subspace $\Rightarrow \text{Span}\{S\} \subset Y$.

□



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(a) $\Rightarrow$ (b). Suppose $x \in V+W$ is written as

$$x = v_1 + w_1$$

and as

$$x = v_2 + w_2$$

where $v_1, v_2 \in V$ and $w_1, w_2 \in W$.

To show: $v_1 = v_2$ and $w_1 = w_2$

Proof

$$0 = x - x = (v_1 - v_2) + (w_1 - w_2)$$

Because V and W are subspaces,

$v_1 - v_2 \in V$ and $w_1 - w_2 \in W$. Because

$$v_1 - v_2 = w_2 - w_1 = -(w_1 - w_2)$$

We have $v_1 - v_2 \in W$ and $w_1 - w_2 \in V$.

hence $(v_1 - v_2) \in V \cap W$ and $(w_1 - w_2) \in V \cap W$. (10)

However, $V \cap W = \{0\}$, and thus $v_1 - v_2 = 0$

and $w_1 - w_2 = 0$, which is what we needed to show.

□

In order to prove $(b) \Rightarrow (a)$, we will prove that $\sim(a) \Rightarrow \sim(b)$.

$\sim(a) \Leftrightarrow \exists u \in V \cap W$ with $u \neq 0$.

$\sim(b) \Leftrightarrow \exists x \in V + W$ that can be expressed in at least two different ways (i.e., NOT UNIQUE).

Because $x \in V + W$, $\exists v \in V$ and $w \in W$ such that $x = v + w$. We ~~also~~ have that

$$x = v + w + u - u = (v + u) + (w - u)$$

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also holds. Moreover, because $u \in V \cap W$, and because V and W are subspaces, we have that $v+u \in V$ and $w-u \in W$. Because $u \neq 0$, we have that

$$V \neq V+u \text{ and } W \neq W-u.$$

So $x = v+w$ and $x = (v+u) + (w-u)$ are distinct ways of expressing x as a sum of elements from V and W , and thus the representation is not unique.

$$4.6 \quad v^1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \quad v^2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \quad v^3 = \begin{bmatrix} 2 \\ 8 \\ -4 \\ 8 \end{bmatrix}$$

$$v^4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad v^5 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ 6 \end{bmatrix}$$

- $\{v^1, v^2\}$ linearly indep
- ~~v^3~~ $v^3 = 4v^1 - 2v^2$
- $\{v^1, v^2, v^4\}$ linearly indep
- $v^5 = v^1 + v^2 + v^4$
- $\text{span}\{v^1, v^2, v^3, v^4, v^5\} = \text{span}\{v^1, v^2, v^4\}$
and $\{v^1, v^2, v^4\}$ is linearly independent.
- dimension is 3.

□

7. We write

$$x = \begin{bmatrix} 8 \\ 7 \\ 4 \end{bmatrix} = \alpha_1 u_{1s} + \alpha_2 u_{2s} + \alpha_3 u_{3s}$$

and solve for the coefficients. Hence,

$$\begin{bmatrix} 8 \\ 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 8 \\ 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 2 \\ -3 \end{bmatrix}$$

$${}_{00}^6 [x]_u = \begin{bmatrix} 9 \\ 2 \\ -3 \end{bmatrix}$$

8.

$$P = [P_1 \mid P_2 \mid P_3]$$

where $P_i = [e_i]_u$

$$\bar{P} = [\bar{P}_1 \mid \bar{P}_2 \mid \bar{P}_3]$$

where $\bar{P}_i = [u_{is}]_{\{e_1, e_2, e_3\}} = [u_{is}]_e$

We compute

$$\bar{P}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \bar{P}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad \bar{P}_3 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

and thus $P = \bar{P}^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

Remark

$$[x]_u = P[x]_e = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 2 \\ -3 \end{bmatrix},$$

which is consistent with our answer in Prob. 7. $\square$

9 (Do not turn in)

(a) We need to show that $\{M_1, M_2, M_3, M_4\}$ is linearly independent. Hence, consider

$$\alpha_1 M_1 + \alpha_2 M_2 + \alpha_3 M_3 + \alpha_4 M_4 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Then

$$\begin{bmatrix} 0 & \alpha_1 \\ \alpha_1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -\alpha_2 \\ \alpha_2 & 0 \end{bmatrix} + \begin{bmatrix} \alpha_3 & 0 \\ 0 & \alpha_3 \end{bmatrix} + \begin{bmatrix} \alpha_4 & 0 \\ 0 & -\alpha_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence, we have

$$\alpha_3 + \alpha_4 = 0$$

$$\alpha_1 - \alpha_2 = 0$$

$$\alpha_1 + \alpha_2 = 0$$

$$\alpha_3 - \alpha_4 = 0$$

That is

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Because both matrices are nonsingular,
we deduce that the only solution is
 $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$, and thus the
set is linearly independent.

(b) We write

$$A = \alpha_1 M_1 + \alpha_2 M_2 + \alpha_3 M_3 + \alpha_4 M_4$$

Using our work from part (a),
we deduce that

$$\begin{aligned}\alpha_3 + \alpha_4 &= 1 \\ \alpha_1 - \alpha_2 &= 2 \\ \alpha_1 + \alpha_2 &= 3 \\ \alpha_3 - \alpha_4 &= 4\end{aligned}$$

Hence

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

which yields

$$\alpha_1 = 2\frac{1}{2}$$

$$\alpha_2 = \frac{1}{2}$$

$$\alpha_3 = 2\frac{1}{2}$$

$$\alpha_4 = -1\frac{1}{2}$$

That is,

$$[A]_M = \begin{bmatrix} 2.5 \\ 0.5 \\ 2.5 \\ -1.5 \end{bmatrix}$$

(c) What is the change of basis